A Redundantly Actuated Hydraulic Ball-on-Plate System: Horizontal-Coordinate Control and a Comparison of Touch-Panel, Monocular-Camera and In-Ball Inertial Position Estimation

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Abstract. Purpose: Ball-on-plate rigs are a standard benchmark for underactuated control, but almost all published systems tilt a small plate with two servomotors and sense the ball with a touch panel or a camera. This paper studies a heavier, industrial-style variant: a round plate carried by four servo-valve hydraulic cylinders on a passive centre post, and compares three ball-position sensing strategies on it. **Methods:** A physics model of the complete rig (closed kinematic chains, servo-valve dynamics with spool lag, overlap and pressure-limited force, sensor noise and bias) was built in MuJoCo and wrapped in a ROS 2 node graph. Only classical methods are used: inverse/least-squares forward kinematics, PID and LQR outer loops, Kalman filtering, a Mahony attitude filter and rolling-constraint odometry. Ball position is obtained (1) from a touch panel, (2) from one overhead camera by back-projecting the detected blob onto the plate plane reconstructed from the cylinder LVDTs, and (3) from an accelerometer–gyroscope–magnetometer unit inside the ball. **Results:** Because the ball centre lies 72 mm above the tilt pivot, the rolling-ball equations have a right-half-plane zero at 13.2 rad/s in the plate-frame position; controlling in a horizontal coordinate removes it. With plate-frame feedback the LQR loop entered a limit cycle (5.82° RMS tilt), whereas horizontal-coordinate feedback brought it to 0.35°. Over 26 Monte-Carlo runs the ball was never lost; circle-tracking RMS error was 2.6, 3.4 and 5.4 mm (LQR) for the three sensing models, with mean RMS estimation errors of 0.9, 3.2 and 5.5 mm. A saddle-mode force-equalisation loop reduced internal cylinder force fight caused by sub-millimetre LVDT offsets from 37.7 N to 4.6 N RMS. **Conclusion:** All three sensing routes are viable in simulation; the in-ball IMU, which needs no external infrastructure, keeps dead-reckoning drift within a few millimetres over the 24 s benchmark but is not absolutely observable. Code, models and data are released for reproduction.

Keywords: ball and plate, hydraulic servo system, redundant actuation, non-minimum phase, LQR, Kalman filter, sensor fusion, ROS 2, MuJoCo

1 Introduction

The ball-on-plate system—a ball rolling freely on a plate whose inclination is actively controlled—is the two-dimensional generalisation of the ball-and-beam and one of the most widely used benchmarks for underactuated, open-loop unstable control [1–4]. Its appeal lies in fast, double-integrator dynamics, strong actuator and sensor limitations, and an immediately visible outcome: the ball either stays on the plate or it does not.

Most published rigs are tabletop devices: a light plate tilted by two rotary servomotors, with the ball sensed by a resistive touch panel [1] or an overhead camera [3]. Larger tilting tables—motion platforms, simulators, test benches—are instead typically driven by linear hydraulic actuators in parallel arrangements [5–8]. Combining the two raises questions that the tabletop literature does not address: (i) hydraulic axes add valve dynamics, pressure-limited force and dead band to the tilt loop; (ii) using more cylinders than platform degrees of freedom (DOF) makes the mechanism over-actuated, so that small inconsistencies produce internal force fight [9]; and (iii) on a physically large plate the pivot generally lies well below the ball, which—as shown in Sect. 3.3—introduces a non-minimum-phase coupling between plate rotation and the ball position measured on the plate.

A second, practical question concerns sensing. A touch panel is accurate but requires a special plate surface; a camera needs a mast and calibration and adds latency; an inertial measurement unit (IMU) inside the ball needs no external infrastructure at all but cannot observe position directly. To the author's knowledge, these three routes have not been compared on the same plant and controller.

This paper presents a complete, reproducible simulation study of such a system. The contributions are:

1. A physics model of a round plate on four servo-valve hydraulic cylinders and a passive centre post, including closed kinematic loops, spool dynamics, overlap, pressure-limited force and LVDT noise, together with the

inverse and least-squares forward kinematics of the over-actuated mechanism and a saddle-mode force-equalisation loop (Sect. 3).

2. A derivation of the rolling-ball dynamics for a pivot offset below the ball, showing a right-half-plane (RHP) zero in the plate-frame position, and a horizontal-coordinate formulation that removes it; its effect is quantified in an ablation (Sects. 3.3, 7.4).
3. Three ball-position estimators sharing one model-based Kalman filter: touch panel, single overhead camera with plate-plane back-projection, and an in-ball accelerometer/gyroscope/magnetometer fusion based on a Mahony attitude filter and rolling-constraint odometry (Sect. 5).
4. A ROS 2 implementation and a Monte-Carlo evaluation of all estimator/controller combinations, with ablations on coordinates, force equalisation, transport delay and magnetometer handling (Sects. 6, 7).

No learning-based components are used anywhere; every element is a classical model-based method. All results are simulation results; Sect. 8 discusses what this does and does not establish.

2 Related work

Ball-on-plate systems. Awtar et al. [1] describe the mechatronic design of a two-motor ball-on-plate system with touch-screen sensing, and Moarref et al. [2] a novel mechanism with position control. Ho et al. [3] implement visual-servoing tracking control with a camera as the only position sensor and validate it experimentally. The non-minimum-phase character of the closely related ball-and-beam—which arises when the ball's rotational inertia and the beam's angular acceleration are retained in the model—was analysed by Hauser et al. [4], who developed an approximate input–output linearisation to cope with it. In contrast to these works, the plant considered here is hydraulically actuated and over-actuated, and the ball's centre lies well above the tilt pivot.

Parallel hydraulic platforms. Platforms driven by linear actuators in parallel go back to Stewart [5]; Dasgupta and Mruthyunjaya [6] and Merlet [7] review their kinematics and design. Electrohydraulic servo axes are modelled in detail by Merritt [10] and Jelali and Kroll [11]; Davliakos and Papadopoulos [8] present model-based control for an electrohydraulic Stewart–Gough platform. Redundantly actuated parallel mechanisms exhibit internal forces that do not move the platform; Müller [9] uses them deliberately for preload control. Here the opposite goal—keeping the redundant internal force near zero—is pursued with a simple equalisation loop.

Estimation. The Kalman filter [12] underlies all three estimators; delayed camera measurements relate to the out-of-sequence-measurement problem studied by Bar-Shalom [13]. Attitude estimation from inertial and magnetic sensors is commonly done with complementary filters on $SO(3)$ [14] or gradient-descent filters [15], initialised e.g. with the TRIAD method [16]. Ball-shaped robots, which share the rolling constraint exploited in Model 3, are reviewed by Chase and Pandya [17]. Camera geometry follows Hartley and Zisserman [18].

Tools. The simulation uses MuJoCo [19], chosen for its handling of closed kinematic chains through equality constraints and its fast, accurate contact model, and the software is organised as a ROS 2 graph [20].

3 System description and modelling

3.1 Mechanism and kinematics

Figure 1 shows the rig and Table 1 its main parameters. A round aluminium plate (Ø600 mm) with a rubber top is supported at its centre by a *passive* post consisting of a vertical slide (heave) and a two-axis gimbal (roll ϕ about x, pitch θ about the rolled y axis). The post carries no load; it only removes the horizontal translations and yaw of the plate. Four double-acting hydraulic cylinders at $0^\circ, 90^\circ, 180^\circ, 270^\circ$ connect the base, through universal joints, to spherical rod ends on the underside of the plate. The plate therefore has three DOF, $q = (\phi, \theta, z)$, driven by four actuators: the mechanism is over-actuated by one.

With the plate rotation $R(\phi,\theta) = R_x(\phi)R_y(\theta)$, pivot $p_0 = (0,0,H+z)^\top$, rod-end anchors a_i (plate frame) and base anchors b_i (world frame), the inverse kinematics (IK) gives the pin-to-pin length of cylinder i as

$$L_i(q) = \left\| p_0 + R(\phi,\theta)\, a_i - b_i \right\|, \qquad i = 1,\dots,4. \tag{1}$$

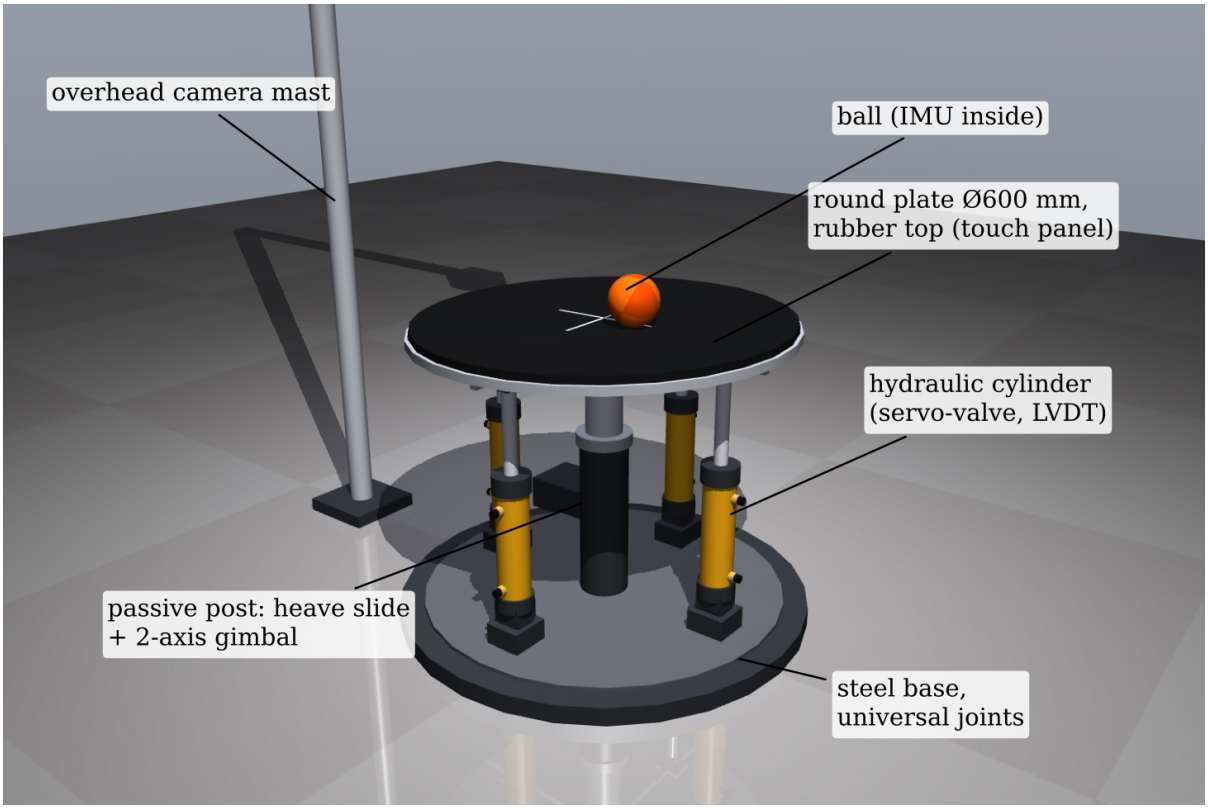


Figure 1: Simulated rig (MuJoCo rendering during circle tracking). The plate is tilted by four servo-valve hydraulic cylinders; a passive centre post fixes the remaining DOF. The overhead camera is used by Model 2, the touch panel (rubber top) by Model 1 and the IMU inside the ball by Model 3.

Because four lengths are measured for three unknowns, the forward kinematics (FK) is solved as the least-squares problem $\hat{q} = \arg\min_q \sum_i (L_i(q) - \tilde{L}_i)^2$ by a warm-started Gauss–Newton iteration (two to four iterations per call). The redundancy is useful: the residual averages LVDT noise, and the plate pose obtained this way is the only plate-state information used by the controllers and estimators.

3.2 Hydraulic axis model and force-fight equalisation

Each cylinder is driven by a proportional servo-valve under a 1 kHz axis controller, modelled after Merritt [10], Jelali and Kroll [11] at the level of detail relevant to the outer loop. With stroke command s^*, its rate $\dot{s}^*$ and LVDT reading $\tilde{s}$:

$$v^* = K_a\,(s^* - \tilde{s}) + k_{\mathrm{ff}}\,\dot{s}^*, \qquad \tau_v \dot{u} = \mathrm{sat}_{[-1,1]}(v^*/v_{\max}) - u, \tag{2}$$

$$u_{\mathrm{eff}} = \mathrm{sgn}(u)\,\frac{\max(|u| - \delta, 0)}{1 - \delta}, \qquad F = \mathrm{sat}_{[-F_{\max}, F_{\max}]}\big(k_v\,(v_{\max} u_{\mathrm{eff}} - \dot{s})\big). \tag{3}$$

Here τ_v is the spool time constant, δ the spool overlap, k_v a velocity gain representing the finite stiffness of the oil column and $F_{\max}$ the pressure-times-area limit. Equation (3) is implemented as a MuJoCo velocity actuator and integrated implicitly, which keeps the stiff oil column numerically stable at a 1 ms time step.

Four stiff position servos on three DOF are over-constrained. Linearising (1) about the level pose shows that the stroke pattern $s = (+1, -1, +1, -1)^\top$ (a "saddle" that would twist the plate) lies in the null space of the platform motion; any component of the four commands, or of LVDT offsets, along s cannot be realised and is instead converted into internal force. A slow integral loop on the measured cylinder forces (pressure sensors on a real rig) removes it:

$$\dot{\Delta} = -k_{\mathrm{eq}}\,\tfrac{1}{4}\,s^\top F, \qquad s^* \leftarrow s^* + \Delta s, \tag{4}$$

with k_{eq} chosen for a bandwidth of roughly 5 rad/s, well below the axis loop.

Table 1: Main model parameters.

Quantity	Value
Plate radius / thickness / mass	300 mm / 22 mm / 4.0 kg
Pivot height (mid stroke)	470 mm
Cylinder anchor radius (base, plate)	220 mm, 220 mm
Cylinder pin-to-pin length / stroke	400 mm / $\pm$75 mm
Ball radius / mass	40 mm / 250 g
Ball-centre height above pivot h	72 mm
Max. piston speed $v_{\max}$	0.30 m/s
Force limit (pressure $\times$ area)	1800 N
Oil-column stiffness k_v	6000 N s/m
Spool time constant / overlap	12 ms / 2 %
Axis position gain K_a / feed-forward	35 s^{-1} / 1.0
LVDT noise (1σ)	20 μm
Tilt limit / tilt-rate limit	$\pm 8^\circ$ / 90°/s
Physics / axis / outer-loop rate	1000 / 1000 / 100 Hz

3.3 Ball on a tilting plate with a pivot offset

Consider one axis. Let x be the ball-centre coordinate along the plate, θ the plate angle (positive θ accelerates the ball towards $+x$), h the height of the ball centre above the pivot axis, r the ball radius, m its mass and J its moment of inertia. With plate unit vectors e_1, e_3, the ball centre is $c = xe_1 + he_3$, so that $\dot{c} = (\dot{x} + h\dot{\theta})e_1 - x\dot{\theta}\,e_3$. Rolling without slipping on the plate gives the ball's absolute angular rate $\omega = \dot{\theta} - \dot{x}/r$. The Lagrangian $\mathscr{L} = \frac{m}{2}\left[(\dot{x} + h\dot{\theta})^2 + x^2\dot{\theta}^2\right] + \frac{J}{2}(\dot{\theta} - \dot{x}/r)^2 - mg(h\cos\theta - x\sin\theta)$ yields

$$\left(m + \frac{J}{r^2}\right)\ddot{x} = mg\sin\theta + mx\dot{\theta}^2 - \left(mh - \frac{J}{r}\right)\ddot{\theta}, \tag{5}$$

and for a solid sphere ($J = \frac{2}{5}mr^2$)

$$\ddot{x} = \tfrac{5}{7}\left[g\sin\theta + x\dot{\theta}^2 - \left(h - \tfrac{2}{5}r\right)\ddot{\theta}\right]. \tag{6}$$

Linearising about rest gives the transfer function from plate angle to plate-frame position

$$\frac{X(s)}{\Theta(s)} = \frac{5}{7}\,\frac{g - (h - \frac{2}{5}r)\,s^2}{s^2}, \tag{7}$$

which, for $h > \frac{2}{5}r$, has a real zero pair at $\pm\sqrt{g/(h - \frac{2}{5}r)}$. With the present geometry ($h - \frac{2}{5}r = 56\,\text{mm}$) the right-half-plane zero lies at 13.2 rad/s, only a few times above the desired outer-loop bandwidth, which limits the achievable performance [21]. Physically, tilting the plate first drags the contact point the "wrong" way before gravity accelerates the ball. Classical ball-and-plate models implicitly assume $h \approx \frac{2}{5}r$ and miss this effect.

The horizontal coordinate $x_h = x + h\sin\theta$, i.e. the horizontal distance of the ball centre from the pivot axis, behaves very differently. From (6),

$$\ddot{x}_h \approx \tfrac{5}{7}g\,\theta + \tfrac{2}{7}(h + r)\,\ddot{\theta}, \qquad \frac{X_h(s)}{\Theta(s)} = \frac{5}{7}\,\frac{g + \frac{2}{5}(h + r)\,s^2}{s^2}, \tag{8}$$

whose zeros are purely imaginary at $\pm$j 14.8 rad/s: there is no RHP zero, and below that frequency the plant is the double integrator $\ddot{x}_h = c\sin\theta$ with $c = \frac{5}{7}g = 7.01\ \text{m/s}^2$. All estimators and controllers in this work therefore operate in horizontal coordinates; plate-frame quantities are converted with $x_h = x + h\sin\theta$ using the LVDT-based tilt.

3.4 Simulation model

The rig is modelled in MuJoCo [19]. Each cylinder is a chain of two hinges (universal joint) and a prismatic joint; the rod end is attached to the plate through a point-equality constraint, closing four kinematic loops. Contacts are enabled only between ball, plate and floor (elliptic friction cone, sliding friction 0.9, small torsional and rolling

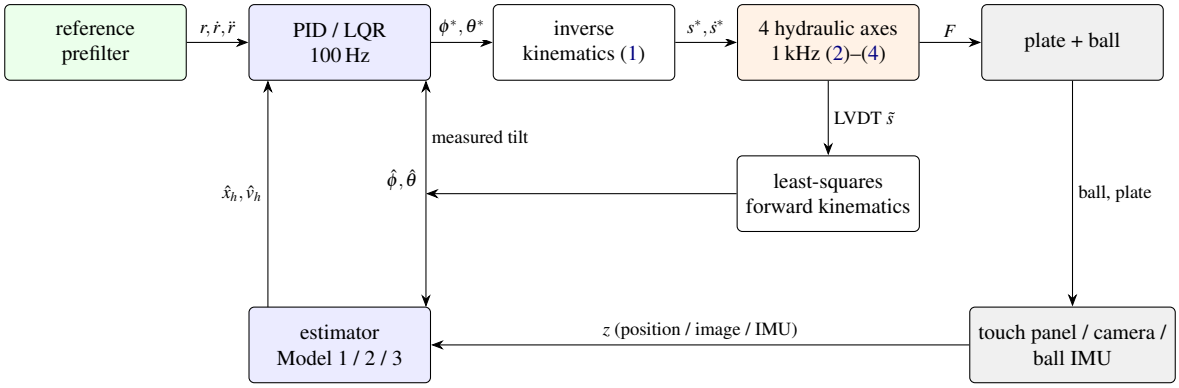


Figure 2: Control structure. The outer loop works in horizontal coordinates; the LS forward kinematics of the four LVDT readings provides the plate tilt used by the controller (LQR state), by every estimator (KF input, coordinate conversion) and, in Model 2, by the back-projection.

friction). An initial rolling-friction value of 0.4 mm—equivalent to 1 % Coulomb drag for a 40 mm ball—caused stiction-induced limit cycles and was reduced to 0.04 mm; with it, the measured ball acceleration on a 1° slope matches $\frac{5}{7} g \sin\theta$ to within 1 %. The ball carries an accelerometer, a gyroscope and a magnetometer at its centre; their outputs receive constant turn-on biases (gyro $\sigma = 0.004$ rad/s per axis) and white noise. The physics runs at 1 kHz.

4 Control

Figure 2 shows the cascade. Each horizontal axis is controlled independently with the virtual input $u_x = \theta$, $u_y = -\phi$ (pitch accelerates the ball along x, roll along $-y$).

Reference prefilter. Targets (step set-points and a circle of 80 mm radius with a 4 s period) are passed through a critically damped second-order filter ($\omega_n = 3.5$ rad/s) that supplies a consistent $(r, \dot{r}, \ddot{r})$ and hence the feed-forward tilt $u_{\mathrm{ff}} = \arcsin(\ddot{r}/c)$.

PID. The PID law acts on the estimated horizontal position and velocity,

$$a = \ddot{r} + k_p(r - \hat{x}_h) + k_d(\dot{r} - \hat{v}_h) + k_i \int (r - \hat{x}_h)\, dt, \qquad u = \arcsin\big(\mathrm{sat}(a/c)\big), \tag{9}$$

with $k_p = 10.0\,\mathrm{s}^{-2}$, $k_d = 5.6\,\mathrm{s}^{-1}$ and $k_i = 3.0\,\mathrm{s}^{-3}$. Anti-windup uses conditional integration: the integrator is frozen when the output saturates in the direction of the error and when $|r - \hat{x}_h| > 25$ mm, so that pushes do not wind it up [22].

LQR. The tilt loop (IK, hydraulic axes, plate) was identified from step responses (Sect. 7.2) as a first-order lag $\tau = 35$ ms. Per axis, the augmented error state $z = [\int e,\ e,\ \dot{e},\ u_{\mathrm{act}} - u_{\mathrm{ff}}]^\top$ with $e = \hat{x}_h - r$ obeys

$$\dot{z} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & c \\ 0 & 0 & 0 & -1/\tau \end{bmatrix} z + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1/\tau \end{bmatrix} (u - u_{\mathrm{ff}}), \tag{10}$$

which is discretised with a zero-order hold at 100 Hz. The discrete LQR [23] with $Q = \mathrm{diag}(40, 600, 60, 0)$ and $R = 40$ gives $K = [0.95,\ 4.09,\ 1.71,\ 0.36]$ and closed-loop poles at $\{-27.11, -8.29, -3.44, -0.26\}$ rad/s; the control is $u = u_{\mathrm{ff}} - Kz$, where the measured tilt u_{act} comes from the LVDT forward kinematics. A first, more aggressive tuning ($R = 12$) tracked equally well in the single-process loop but produced more plate jitter; in the ROS 2 node graph this jitter caused micro-slip of the ball and roughly tripled the Model 3 estimation error in a development run, so the weights above were chosen as the compromise.

Limits. Both controllers share a magnitude limit of $\pm 8^\circ$ and a rate limit of 90°/s on the commanded tilt. The IK converts (ϕ^*, θ^*) into four stroke commands and their rates, which feed (2).

5 Ball-position estimation

5.1 Common model-based Kalman filter

All three estimators share a two-state Kalman filter [12] per horizontal axis, $\xi = [x_h,\ v_h]^\top$, whose prediction uses the known input $a = c \sin u_{\text{act}}$ from the measured tilt:

$$\xi_{k+1} = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix} \xi_k + \begin{bmatrix} \Delta t^2/2 \\ \Delta t \end{bmatrix} c \sin u_{\text{act},k} + w_k, \tag{11}$$

with w_k an acceleration-noise process (σ_a between 0.25 and 0.35 m/s^2) that also absorbs the neglected $\frac{2}{7}(h+r)\ddot{\theta}$ term of (8). Using the physics as prediction model lets the filter coast through dropped measurements and produces a velocity estimate with far less lag than a differentiator. The models differ only in their measurement updates.

5.2 Model 1: direct position sensing

A touch panel (or motion capture) reports the plate-frame position at 100 Hz with $\sigma = 0.8$ mm. It is converted to $z_h = z + h \sin u_{\text{act}}$ and used as a position update.

5.3 Model 2: single overhead camera

A pinhole camera 0.95 m above the pivot (640×480 pixels, 45° vertical field of view, 60 fps, 12 ms capture-to-processing latency) observes the orange ball on the black plate (Fig. 3). Each frame is converted to HSV, thresholded, cleaned with morphological opening and closing, and the largest blob's centroid (u, v) is computed from image moments. A single camera provides no depth, but the plate pose at the capture time is known from the LVDTs, and the ball centre must lie on the plane parallel to the plate at height h. With camera centre c_{cam}, rotation R_{cam} and intrinsics K, the viewing ray is $d = R_{\text{cam}}\,\text{diag}(1,-1,-1)K^{-1}[u, v, 1]^\top$ and [18]

$$\lambda = \frac{n^\top (p_0 + hn - c_{\text{cam}})}{n^\top d}, \qquad \begin{bmatrix} x \\ y \\ \cdot \end{bmatrix} = R^\top \left(c_{\text{cam}} + \lambda d - p_0 \right), \tag{12}$$

where $n = R e_z$ is the plate normal. Measurements are gated (5σ plus 20 mm) and, because they arrive τ_d after capture, are applied as $z + \hat{v}\tau_d$ with inflated noise—a first-order approximation of the exact out-of-sequence update [13]. The extrinsic calibration carries a seeded error of about 1 mm and 0.1°. Lens distortion and motion blur are not simulated.

5.4 Model 3: IMU inside the ball

An IMU cannot measure position, but a ball rolling without slip turns its rotation into translation. The estimator has three layers.

Attitude. A Mahony complementary filter on $SO(3)$ [14] at 200 Hz integrates the bias-corrected gyro rate with the correction

$$\hat{\omega} = \omega_m - \hat{b} + k_a\, w(\|a_m\|) \left(\bar{a}_m \times \hat{R}^\top \bar{f} \right) + k_m\, e_m, \qquad \dot{\hat{b}} = -k_i\, e, \tag{13}$$

where bars denote unit vectors. Three details matter inside a ball that is pushed and shaken. (i) The reference specific force $f = g e_z + \hat{a}_{\text{lin}}$ includes the model-predicted ball acceleration $c \sin u$. (ii) The accelerometer weight $w = \exp(-((\|a_m\| - g)/0.6)^2)$ falls off during impacts and plate jerks. (iii) The bias integral is enabled only once the attitude error is small, so that a large initial error cannot wind it up. The filter is initialised with TRIAD [16] from ten averaged quiet samples. In the implemented version the magnetometer corrects *heading only*: the horizontal components of the measured and reference fields are compared in the world frame and the correction acts about the vertical axis, $e_m = \hat{R}^\top e_z \left[(\mathrm{h}(\hat{R}\bar{m}) \times \mathrm{h}(\bar{m}_{\text{ref}})) \cdot e_z \right]$, so that magnetic disturbances cannot tilt the attitude estimate. Sect. 7.7 compares this with the conventional full-vector correction [14, 15].

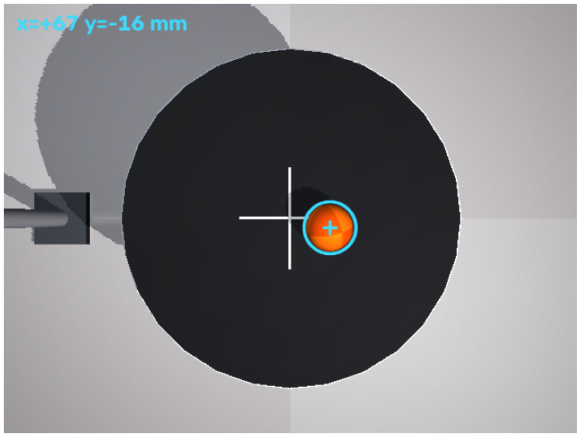


Figure 3: Overhead camera frame during circle tracking with the detected ball (cyan circle), its centroid and the back-projected plate-frame position (Model 2).

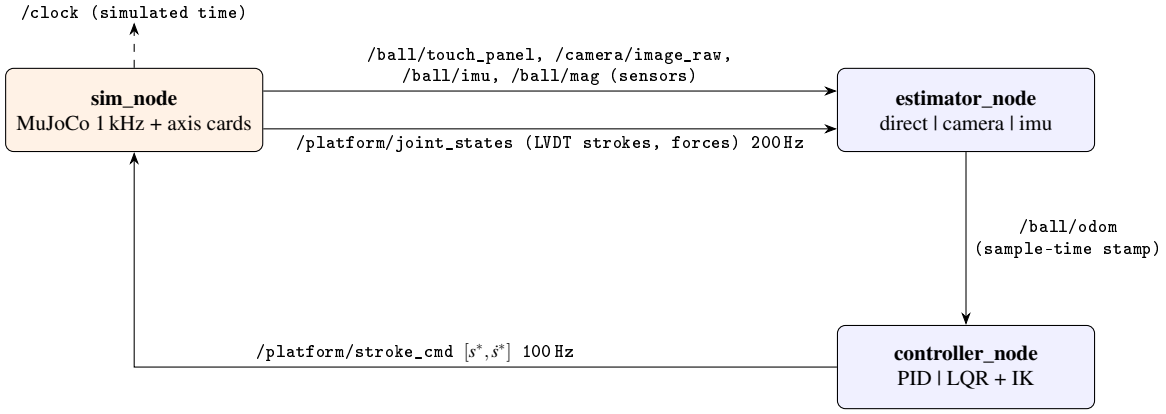


Figure 4: ROS 2 node graph. The simulator owns time (`/clock`); estimator and controller run on simulated time and are event-driven by the measurements, so a slower-than-real-time simulation (e.g. software rendering of camera images) does not change the closed-loop behaviour.

Rolling-constraint odometry. With the plate rotation R and plate angular rate ω_p (from the LVDT poses), the ball velocity relative to the plate, expressed in the plate frame, follows from the no-slip condition at the contact point:

$$v_{\text{rel}} = r\left(R^{\top}\hat{R}(\omega_m - \hat{b}) - \omega_p\right) \times e_z, \qquad v_h = v_{\text{rel}} + h\cos u\,\dot{u}. \tag{14}$$

The spin component about the plate normal drops out of the cross product, which is why heading and attitude errors, rather than gyro noise, dominate the error budget: an attitude error δ leaks the spin ω_n into an apparent rolling rate $\omega_n\delta$.

Filter. The rolling velocity (14) is a velocity measurement ($\sigma = 20$ mm/s) for the shared Kalman filter. Position is obtained by integrating it from a known start point (the ball is placed at a "home" position). It is therefore *not* absolutely observable and drifts with gyro-bias residuals, heading error and wheel-slip during hard pushes.

6 Software architecture

The implementation separates a ROS-free core (models, kinematics, controllers, estimators) from thin ROS 2 [20] nodes (Fig. 4), so that the identical code runs both in a deterministic single-process loop (used for all Monte-Carlo results) and as a distributed node graph. Standard message types are used throughout (`JointState` for LVDT strokes and cylinder forces, `Image`/`CameraInfo`, `Imu`/`MagneticField`, `Odometry` for the estimate).

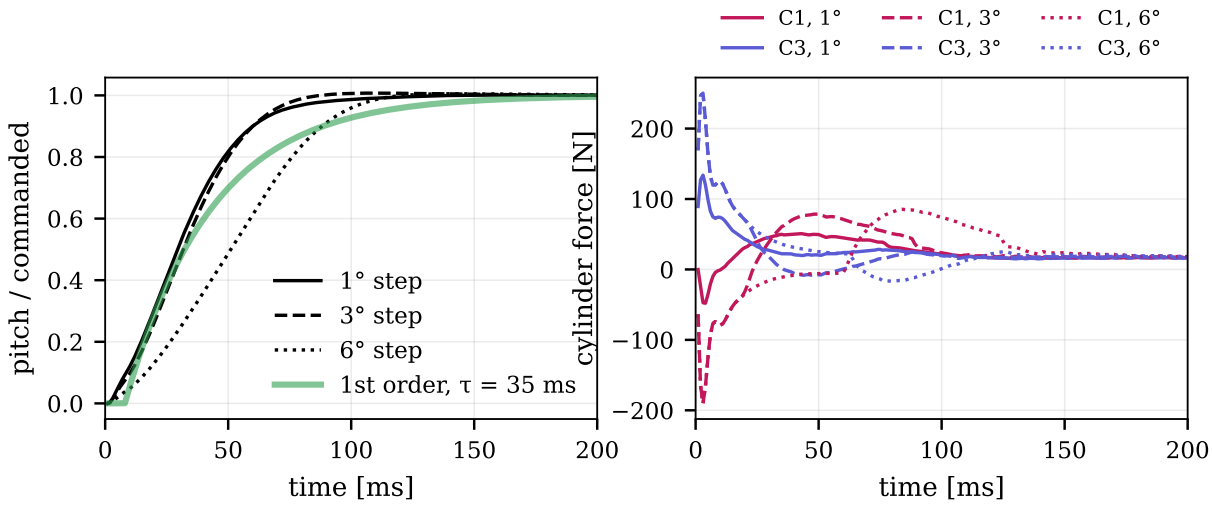


Figure 5: Tilt-loop step responses (left, normalised) and forces in the two cylinders that move in opposite directions (right). The overshoot-free response is captured by a first-order lag; larger steps are rate-limited by the valve.

Two timing details were essential for the node graph to match the single-process loop. First, the estimator stamps its output with the *measurement* time and the controller derives its sample interval from these stamps rather than from the node clock; with `/clock` published every 5 ms, clock-based intervals jittered between 5 and 15 ms and disturbed the integrators. Second, converting floating-point seconds to `builtin_interfaces/Time` by separately truncating the seconds and rounding the nanoseconds maps, e.g., 9.9999999 s to 9 s; the resulting one-second jumps propagated through the Kalman prediction and caused periodic plate jerks (and, with Model 3, loss of the ball) until the conversion was done through integer nanoseconds. The whole stack is packaged in Docker images (ROS 2 Jazzy, MuJoCo, software OpenGL), including a variant that exposes an interactive 3-D view through a browser-based VNC client.

7 Results

7.1 Protocol and metrics

All experiments use the same 24 s scenario: the ball is released at rest at $(110, -70)$ mm and brought to the centre; at $t = 5$ s a horizontal force of 2.2 N is applied for 60 ms (a "push"); from 8 to 16 s the ball tracks two laps of a circle of 80 mm radius; from 16 to 19.5 s it holds the set-point $(-100, 60)$ mm; at 19.5 s a diagonal push of 2.3 N is applied; then the ball returns to the centre. Reported metrics are the RMS distance between true ball and reference during 10–16 s (*circle RMS*, plate frame), the mean distance during 23–24 s (*final hold*), the RMS and maximum distance between true and estimated position (*est.*), and whether the ball left the plate. Monte-Carlo runs vary the random seed, which changes all sensor noise realisations, IMU biases and camera calibration errors.

7.2 Tilt-loop identification

Figure 5 shows the plate pitch after IK step commands. For small steps the response is well described by 9 ms dead time and a first-order lag reaching 63 % at 37 ms and 90 % at 60 ms; for a 6° step the valve saturates and the 63 % time grows to 62 ms. The LQR design model uses $\tau = 35$ ms.

7.3 Overall performance

Table 2 summarises the Monte-Carlo results and the single ROS 2 runs. In none of the 26 runs was the ball lost; the largest cylinder force was 221 N, about 12 % of the 1800 N force limit. Figure 6 shows the ball paths and Fig. 7 the error histories for the three sensing models with LQR.

Model 1 is the reference: the estimation error is of the order of a millimetre except during pushes, which the model-based filter cannot anticipate. Model 2 achieves almost the same tracking performance. Its estimation

Table 2: Monte-Carlo results (mean ± standard deviation over seeds; n runs per row) and one ROS 2 node-graph run per row (seed 0).

Estimator	Ctrl.	n	lost	circle RMS [mm]	final hold [mm]	est. RMS [mm]	est. max [mm]	ROS 2 (seed 0) circle / est. [mm]
Direct	PID	5	0	2.63 ± 0.04	1.9 ± 0.1	0.9 ± 0.0	8.5 ± 0.3	2.88 / 1.6
Direct	LQR	5	0	2.60 ± 0.05	1.3 ± 0.1	0.9 ± 0.0	8.5 ± 0.3	2.96 / 1.7
Camera	PID	3	0	3.88 ± 1.20	2.7 ± 1.5	3.2 ± 1.3	19.1 ± 0.7	3.85 / 3.1
Camera	LQR	3	0	3.42 ± 1.30	2.3 ± 1.7	3.2 ± 1.3	19.1 ± 0.7	3.42 / 3.2
IMU fusion	PID	5	0	5.46 ± 1.36	6.3 ± 3.3	5.4 ± 2.0	9.3 ± 2.6	5.84 / 6.0
IMU fusion	LQR	5	0	5.36 ± 1.91	6.0 ± 3.2	5.5 ± 2.3	9.7 ± 3.0	5.13 / 5.9

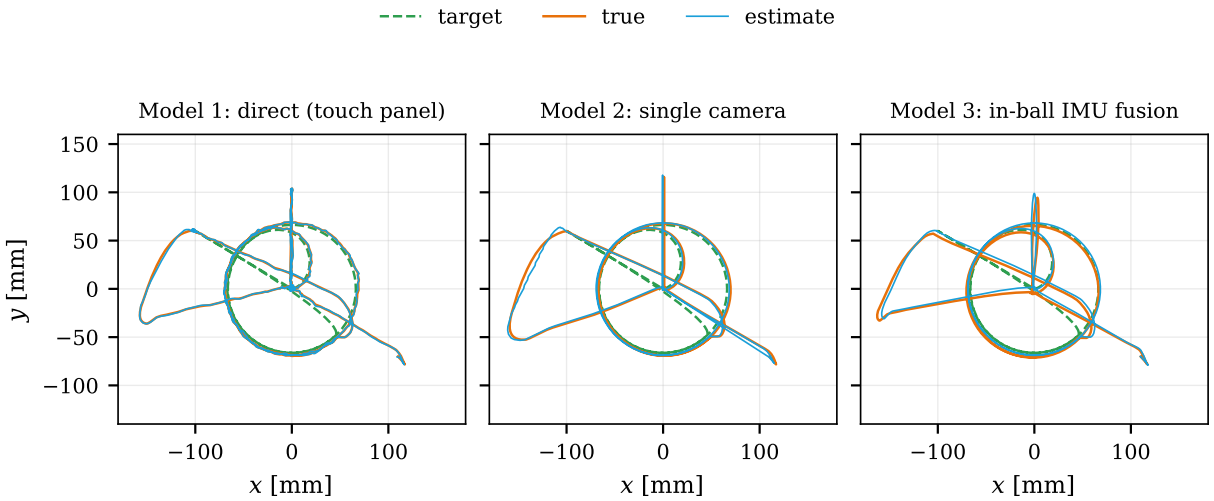


Figure 6: Ball path on the plate (seed 0, LQR): true position, estimate and target. The circle is followed with all three sensing models; the two push responses are the excursions towards the top and bottom left.

error has a near-constant component of 1–4 mm, depending on the realisation of the injected extrinsic calibration error—a bias that no amount of filtering removes—plus short spikes at the pushes, recovered within a few frames. Model 3 tracks the circle with somewhat larger errors. Its estimation error grows slowly and in steps: each push or strong plate acceleration adds a few millimetres through slip and transient attitude error, whereas gyro noise itself is negligible. Over the 24 s benchmark the worst estimation error among the IMU runs was 13.4 mm (LQR).

7.4 Effect of the coordinate choice

Table 3 and Fig. 8 compare estimator and controller operating in plate coordinates (h set to zero in the conversions) with the horizontal coordinates of Sect. 3.3. With plate coordinates the LQR loop enters a sustained oscillation (5.82° RMS tilt between 2 and 5 s, circle error 18.56 mm), consistent with the RHP zero of (7). A plausible mechanism is that the LQR, designed for a pure double integrator, applies a high gain to the velocity estimate, which in plate coordinates contains the spurious, plate-rotation-induced term $-h\dot{\theta}$. The less aggressive PID tuning remains stable in plate coordinates. In horizontal coordinates both controllers are stable and quiet (0.20° and 0.35°).

The table also exposes a subtlety. In its own coordinates the horizontal controller tracks the circle to within 0.87 mm, but in the plate frame the error is 2.58 mm, because during circle tracking the plate is tilted by the feed-forward angle and x_h and x differ by $h \sin u_{\mathrm{ff}} \approx 2$ mm. The plate-coordinate PID, which tracks x directly, reaches 1.08 mm in the plate frame. The two advantages can be combined by keeping horizontal-coordinate feedback but shifting the reference given to the controller,

$$r_h = r + h\ddot{r}/c \;\approx\; r + h \sin u_{\mathrm{ff}}, \tag{15}$$

which reduced the plate-frame circle error to 1.57 mm (LQR) and 2.08 mm (PID) without any loss of stability. This shift was evaluated as an extension and is not part of the baseline results in Table 2.

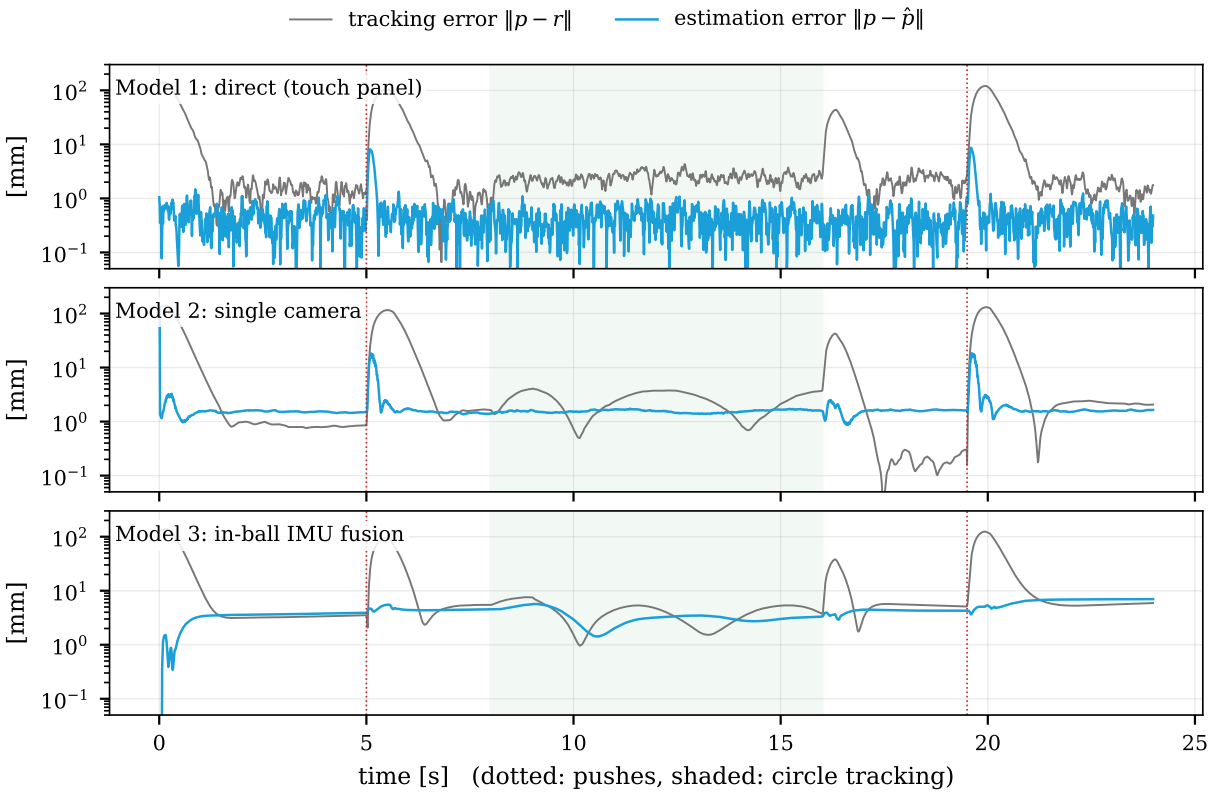


Figure 7: Tracking error and estimation error over time (seed 0, LQR, logarithmic scale). Dotted lines: pushes; shaded: circle tracking.

7.5 Force-fight equalisation

With ideal LVDTs the saddle-mode internal force is small with or without equalisation (4.9 N vs. 4.6 N RMS), because the four IK commands are consistent and LVDT noise is only 20 μm. With per-cylinder zero offsets between 0.1 and 0.3 mm—typical sensor and mounting tolerances—the cylinders fight with 37.7 N RMS; the equalisation loop (4) removes the inconsistency within a fraction of a second and brings it back to 4.6 N (Fig. 9). Tracking is unaffected in both cases (2.56 vs. 2.56 mm), since the fight acts in the null space of the platform motion; its cost is internal load, heat and wear. The large peak forces visible in Fig. 5 are not force fight: they are opposite cylinders accelerating the plate in pitch.

7.6 Robustness to transport delay

Distributing the loop over processes and networks adds latency. Figure 10 shows the circle error when an extra delay is inserted between controller and axis cards (Model 1). Both controllers degrade gracefully: PID from 2.6 to 4.3 mm and LQR from 2.6 to 5.8 mm at 80 ms, without losing the ball. The LQR, which is tuned more aggressively, degrades faster because its design model contains no delay.

7.7 IMU fusion: attitude and drift

Table 4 and Fig. 11 evaluate Model 3 over 20 seeds and compare the implemented heading-only magnetometer correction with the conventional full-vector correction. Attitude accuracy is the same (median errors of 1–2°; paired Wilcoxon $p = 0.90$). The position estimate, however, was better with the full-vector correction (mean worst-case error 7.1 vs. 9.6 mm, $p = 0.021$; mean RMS 4.4 vs. 5.5 mm, $p = 0.070$). In this simulation the magnetometer has only a small constant hard-iron bias, so its tilt information is useful rather than harmful; the heading-only variant would pay off only in the presence of magnetic disturbances that are not simulated here (e.g. the steel base and hydraulic power unit near the plate). Given these results the full-vector correction is the better default for the modelled sensor, and the heading-only variant should be reserved for magnetically disturbed environments.

Table 3: Ablation on the coordinates used by the estimator and controller (Model 1). Plate-frame and horizontal-frame circle errors are both reported because the horizontal controller regulates x_h, which differs from the plate-frame position by $h \sin u$.

Ctrl.	Coordinates used by KF and controller	tilt RMS 2–5 s [deg]	circle RMS plate frame [mm]	circle RMS horiz. frame [mm]	final hold [mm]
PID	plate coordinates	0.50	1.08	1.36	2.24
PID	horizontal coordinates	0.20	2.58	0.90	2.10
PID	horizontal + ref. shift (Eq. 15)	–	2.08	–	1.94
LQR	plate coordinates	5.82	18.56	11.99	14.16
LQR	horizontal coordinates	0.35	2.58	0.87	1.41
LQR	horizontal + ref. shift (Eq. 15)	–	1.57	–	1.28

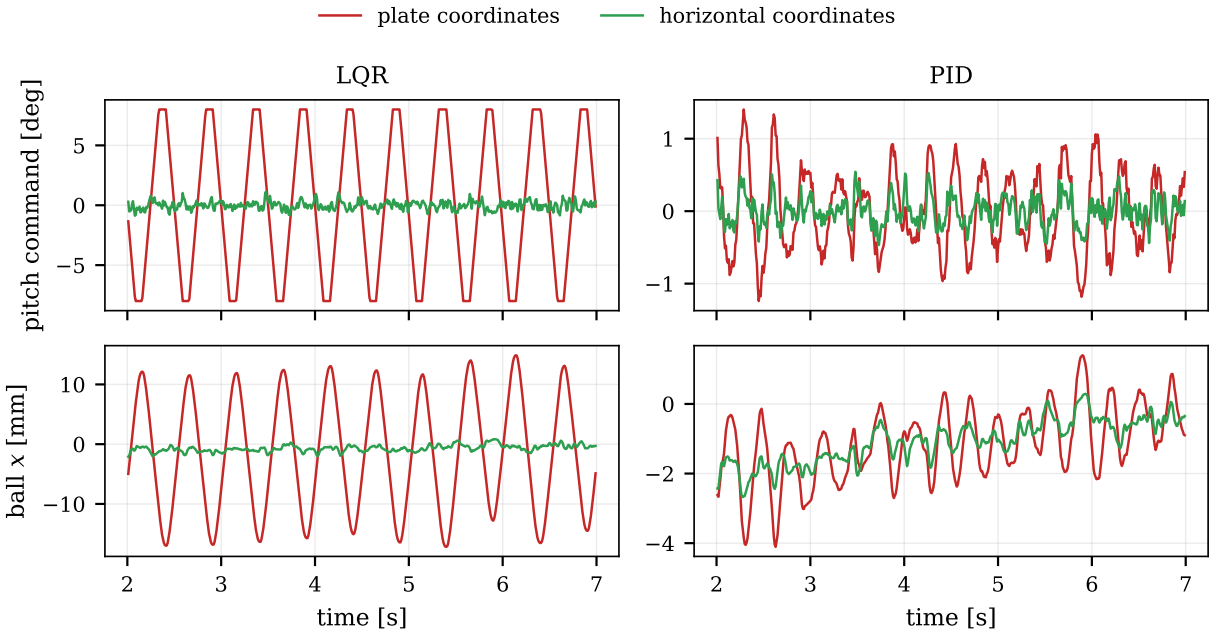


Figure 8: Pitch command and ball x position with plate-frame and horizontal-frame feedback. In plate coordinates the LQR loop oscillates; the PID tuning tolerates it.

7.8 Node graph versus single process

The last column of Table 2 reports one run of each combination through the ROS 2 node graph, with message passing, simulated time and the event-driven timing of Sect. 6. Circle errors and estimation errors are within about 1 mm of the single-process values, i.e. the distributed implementation preserves the closed-loop behaviour; the remaining differences come from the varying message latency.

8 Discussion and limitations

What the results establish. Within the model, a four-cylinder hydraulic ball-on-plate system can be stabilised and made to track with millimetre accuracy by classical controllers, provided that (i) the outer loop works in horizontal coordinates or is detuned enough to tolerate the RHP zero, and (ii) the tilt-loop lag is accounted for. All three sensing routes are adequate for balancing; the camera adds a calibration bias, and the IMU adds slow drift but needs no external infrastructure.

Limitations. All results are from simulation, and several effects are idealised: the hydraulic model has no pressure dynamics, supply-pressure variation, cylinder friction or temperature dependence; MuJoCo's soft-contact model approximates rolling and slip; the camera image is an ideal pinhole rendering without distortion, blur or rolling shutter; the IMU error model contains constant biases and white noise but no temperature drift, scale-factor or misalignment errors, and no magnetic disturbances. The IMU position estimate is unobservable in principle:

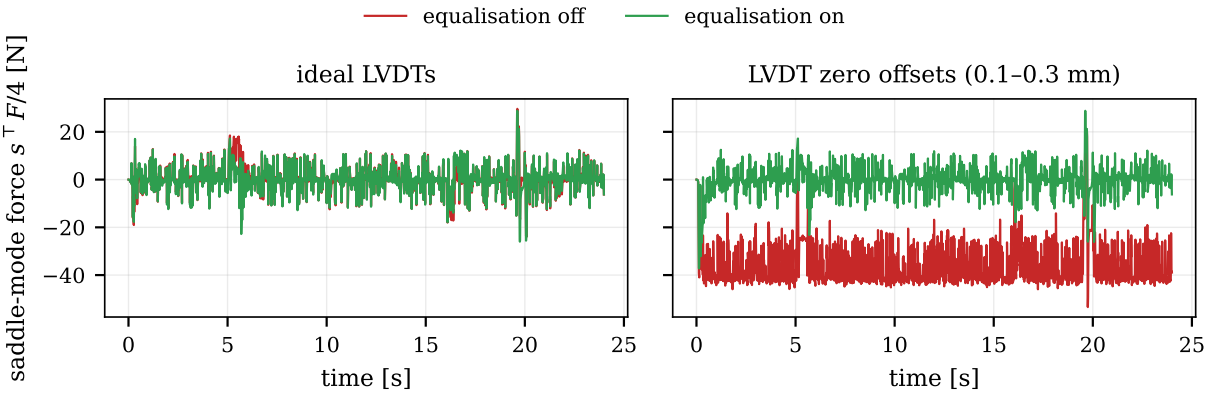


Figure 9: Saddle-mode internal force $\frac{1}{4}s^\top F$ without and with equalisation, for ideal LVDTs (left) and for sub-millimetre LVDT zero offsets (right).

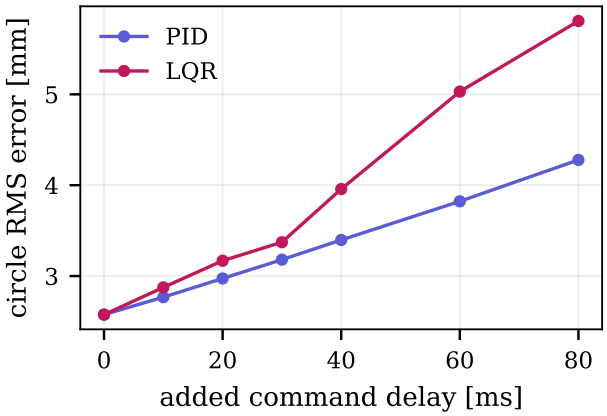


Figure 10: Circle-tracking error versus added command delay (Model 1).

over the 24 s benchmark its drift stays at a few millimetres, but over minutes it will accumulate without bound unless an absolute reference is added. Finally, the controllers were tuned on the model, and the Monte-Carlo study varies sensor realisations, not plant parameters.

Future work. The most direct extensions are hardware validation on a scaled rig, fusing the camera at low rate with the IMU to bound the drift, delay-aware LQR or model-predictive control using the identified tilt dynamics, and adopting the reference shift (15) in the baseline controllers.

9 Conclusion

A round plate on four servo-valve hydraulic cylinders was modelled, controlled and evaluated with three ball-position sensing strategies using only classical methods. The pivot offset under the ball introduces a right-half-plane zero at 13.2 rad/s in the plate-frame position; controlling in a horizontal coordinate removes it, and a small reference shift recovers plate-frame accuracy. A saddle-mode equalisation loop removes the internal force fight that the over-actuation creates under realistic sensor offsets. Touch-panel, single-camera and in-ball IMU sensing all balance the ball in every Monte-Carlo run, with circle errors of a few millimetres; their differences lie in bias (camera calibration) and drift (IMU dead reckoning) rather than in noise. The complete code, models, data and figure scripts are released so that every number in this paper can be regenerated.

Acknowledgements [To be completed by the author.]

Table 4: Model 3 (LQR) with heading-only (implemented) and full-vector magnetometer correction over n seeds: attitude error (time mean), position estimation error and circle tracking error (mean ± s.d.).

Magnetometer correction	n	lost	attitude err. [deg]	est. RMS [mm]	est. max [mm]	worst [mm]	circle RMS [mm]
heading-only (implemented)	20	0	1.57 ± 0.45	5.5 ± 1.9	9.6 ± 3.2	17.8	5.0 ± 1.7
full-vector	20	0	1.65 ± 0.77	4.4 ± 2.4	7.1 ± 3.2	17.4	5.4 ± 2.4

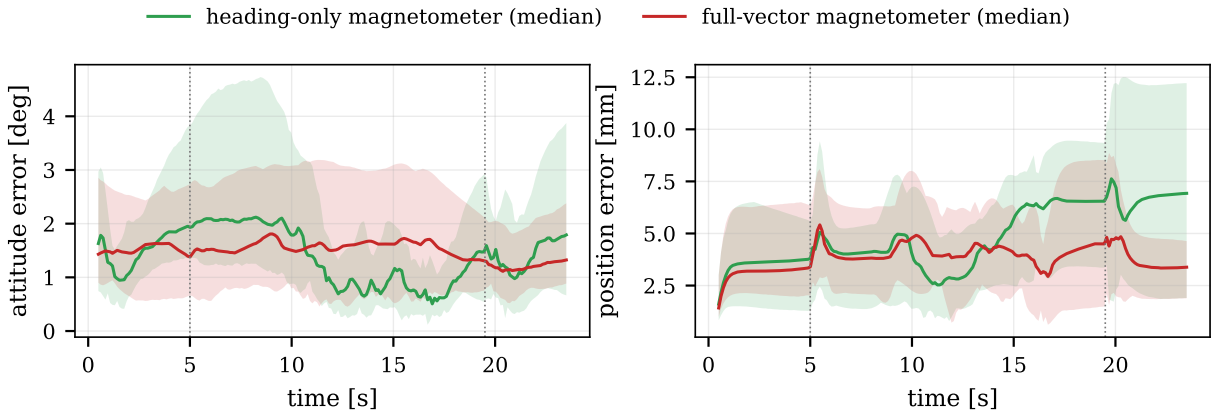


Figure 11: Model 3 attitude error (left) and position error (right) over time: median and 10–90 % band over seeds. Dotted lines: pushes.

Declarations

- **Funding:** [To be completed by the author.]
- **Competing interests:** [To be completed by the author.]
- **Data and code availability:** The simulation code, ROS 2 package, Docker files, experiment and figure scripts, and the raw data behind every table and figure are contained in the project repository [public URL to be added on release].
- **Use of AI tools:** [To be reviewed by the author.] The simulation software and a first draft of this manuscript were prepared with the assistance of an AI system (Claude, Anthropic). All results were generated by the released code, and the author has reviewed and takes responsibility for the content.
- **Author contributions:** [To be completed by the author.]

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